Carnegie Mellon University
Master of
Software Engineering

Layered style

LG Architecture Training Program
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Agenda

- Module views
  - View showing module decomposition
  - View showing module use
  - Layered style
  - MVC style
- Data model/domain model
- Hexagonal architecture
- Microkernel pattern

Layered style

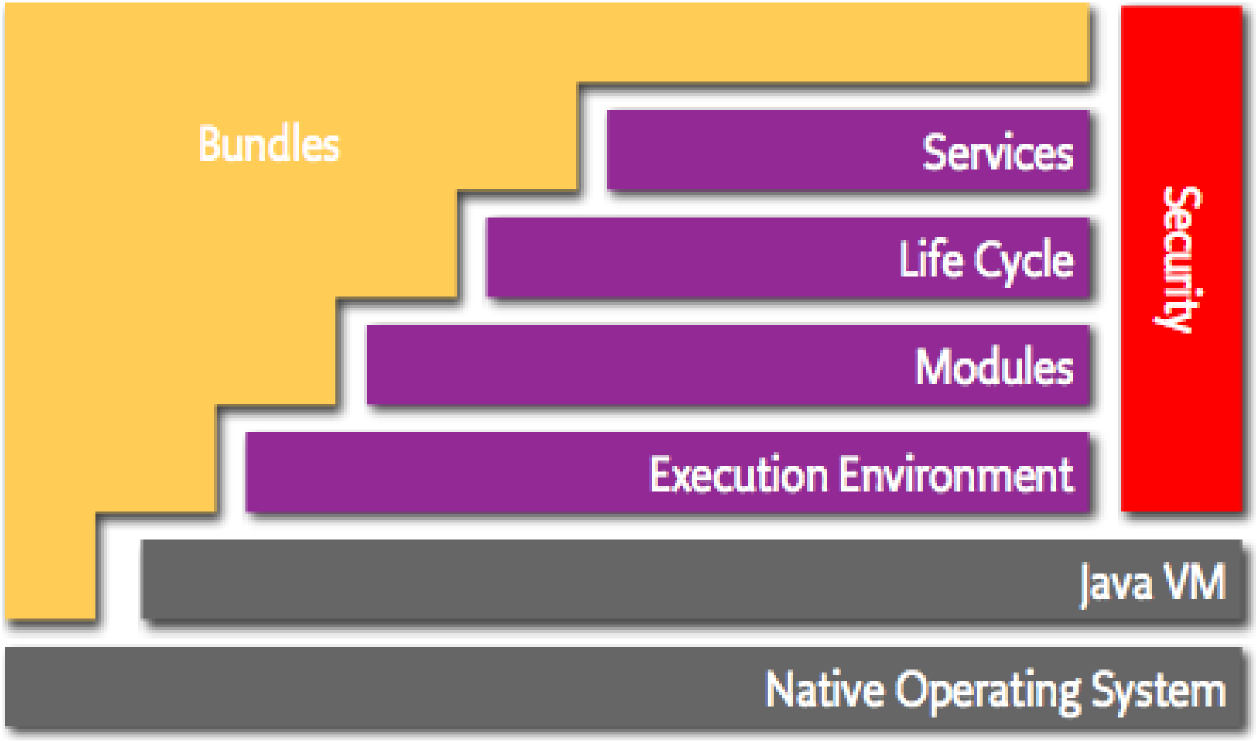
- **Element**: layer

> *A layer is a grouping of modules that together offer a cohesive set of services to other layers.*
> [Clements10]

- **Relation**: “allowed to use” (or simply “can use”)
- **Constraint**: the allowed-to-use relation must be unidirectional

It’s common and useful to describe the organization of modules in layers. However, many layered views are not well documented.

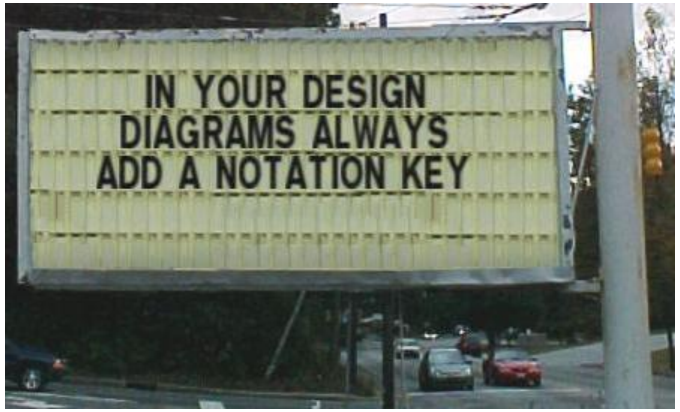
Layered architecture of the OSGi framework

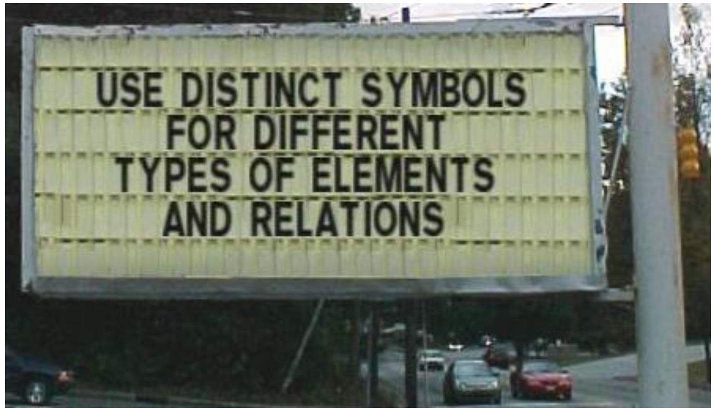


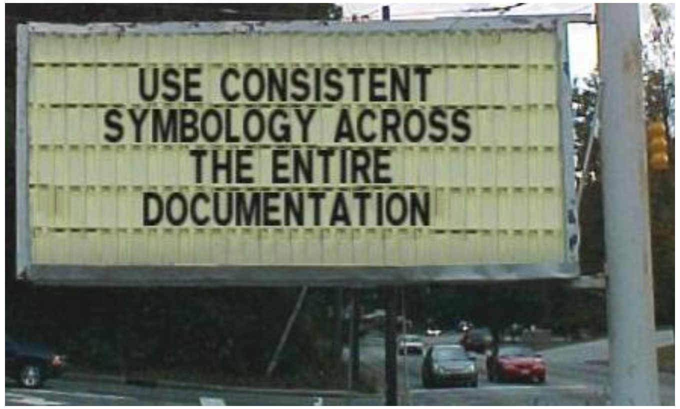
The notation in this diagram has at least two ambiguities

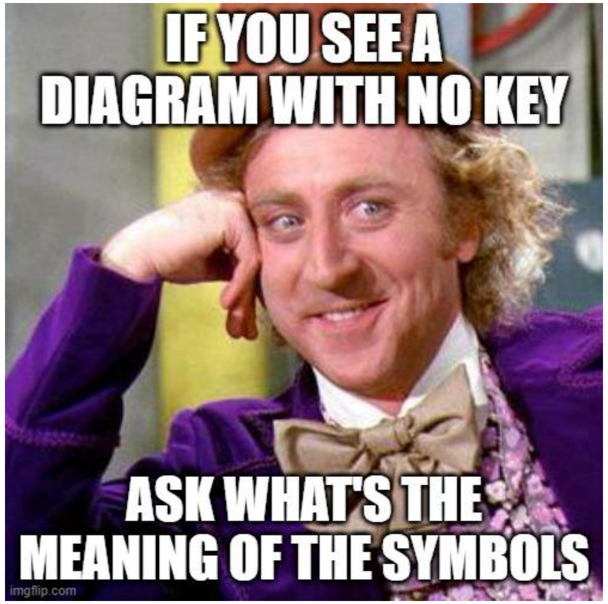
Source: www.osgi.org/resources/architecture/

(Design diagram tips)



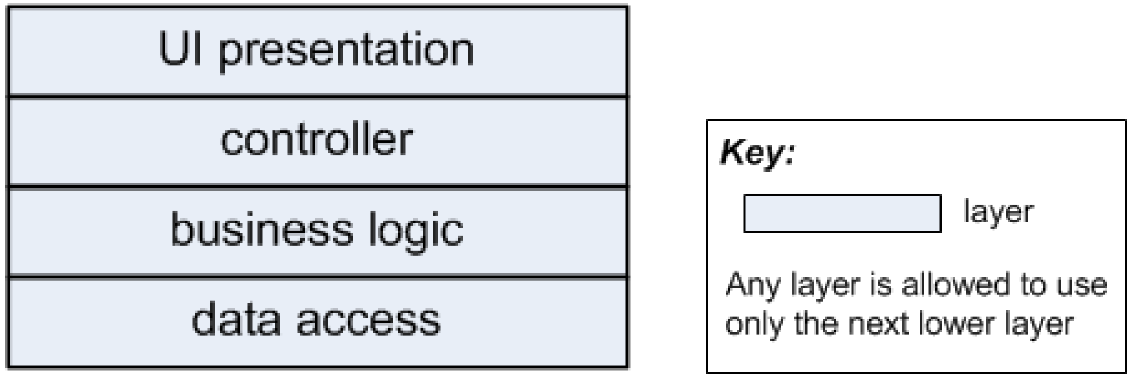






Layered style and modifiability

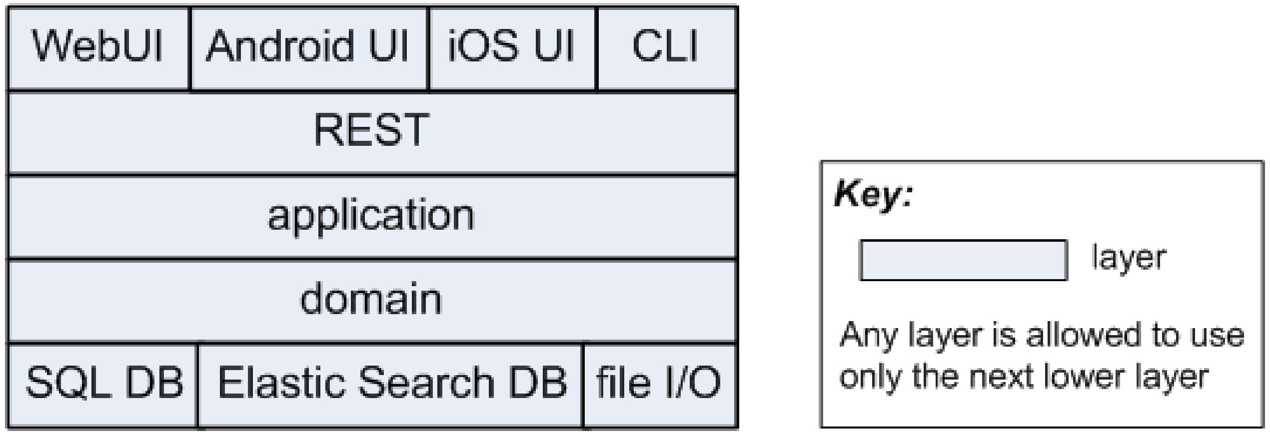
- Layers can be used for “separation of concerns.” Example:



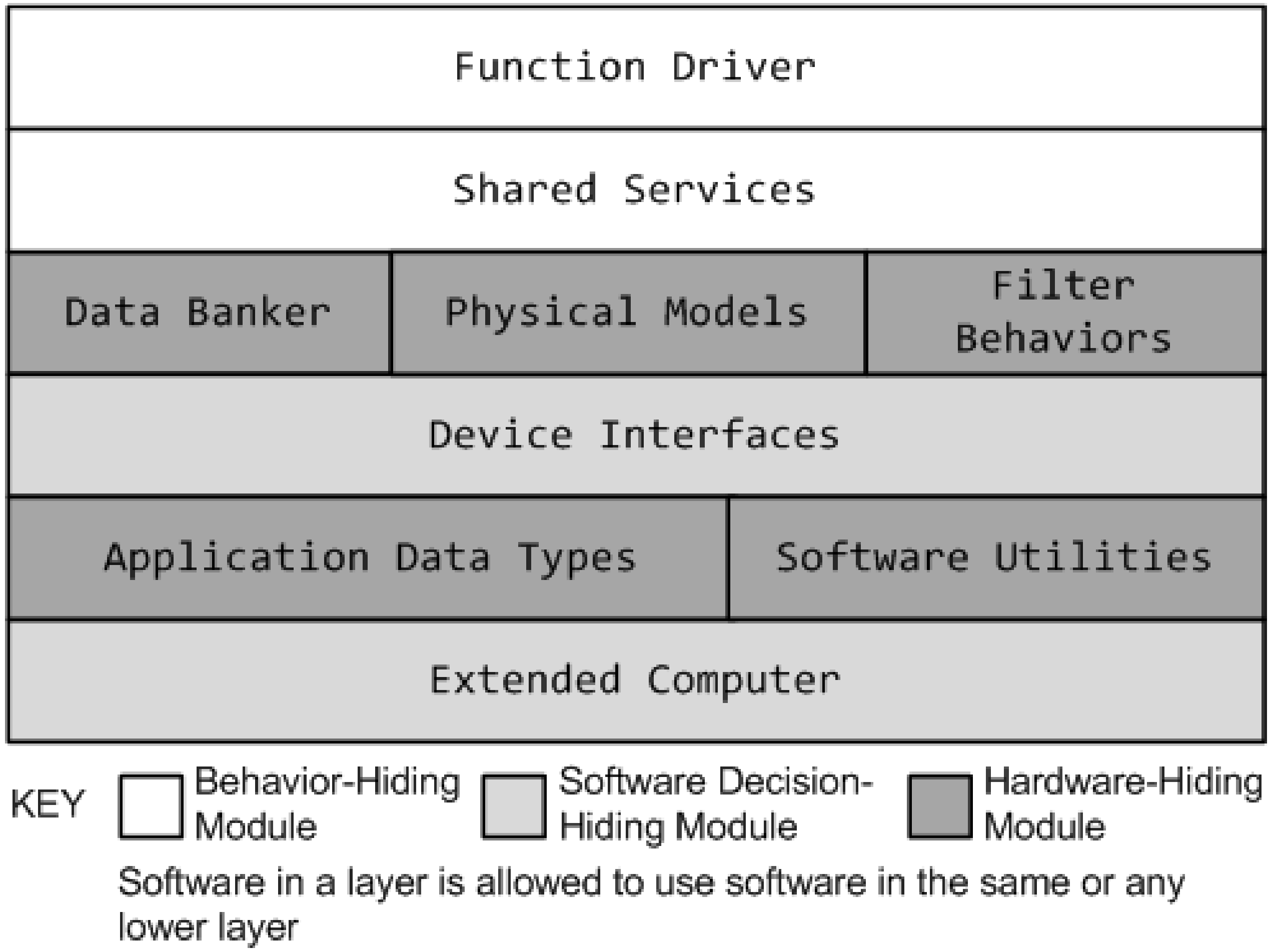
- A layer is a logical grouping that is normally not clearly visible in the source code
- The source code reveals the actual usage dependencies, which are bound by the allowed-to-use dependencies of the layered design

Layered style and portability

- Layers can be used to isolate technology-specific code
- The goal is to make the system more portable
- Example:

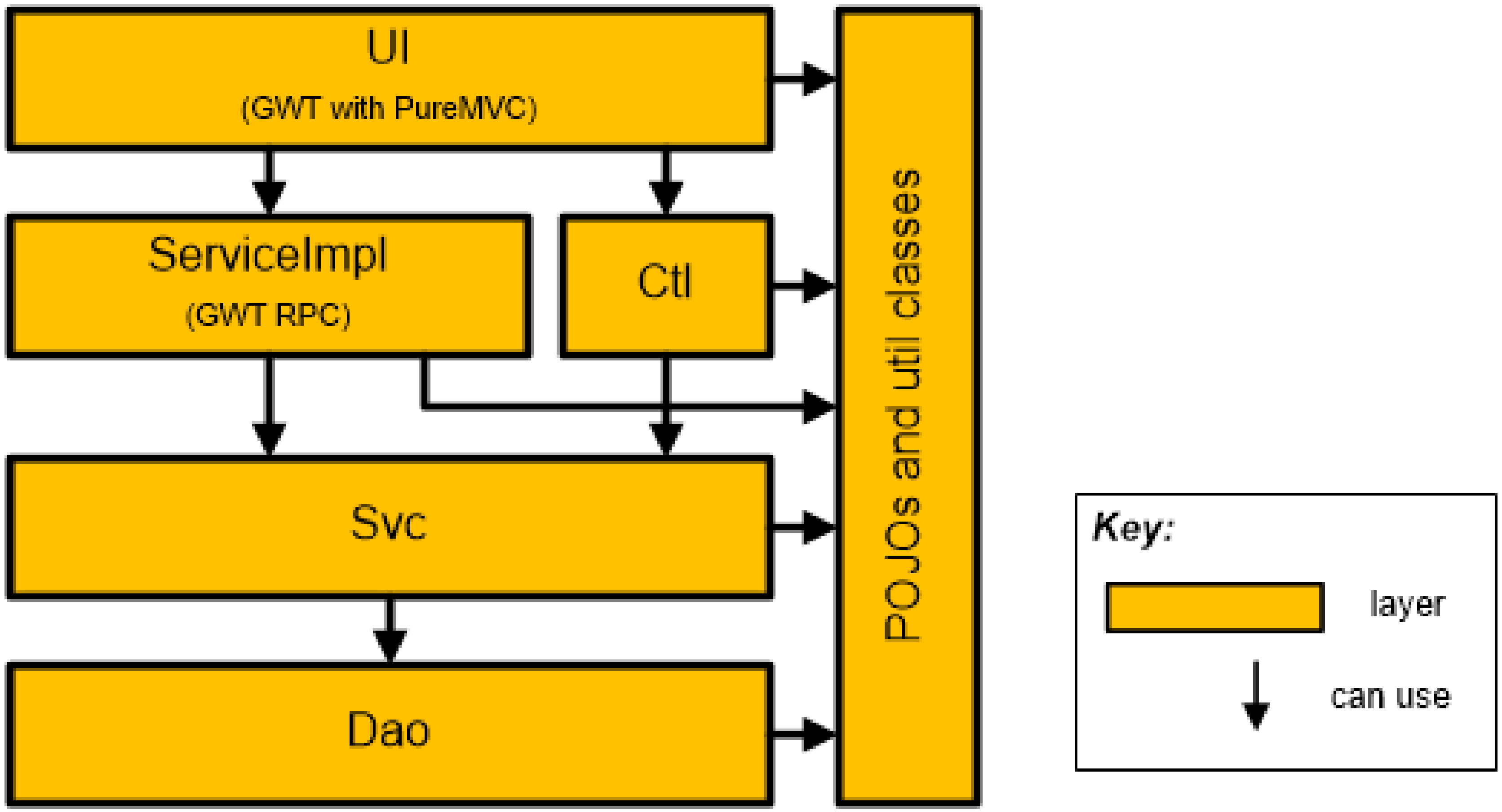


Layered style – example 1

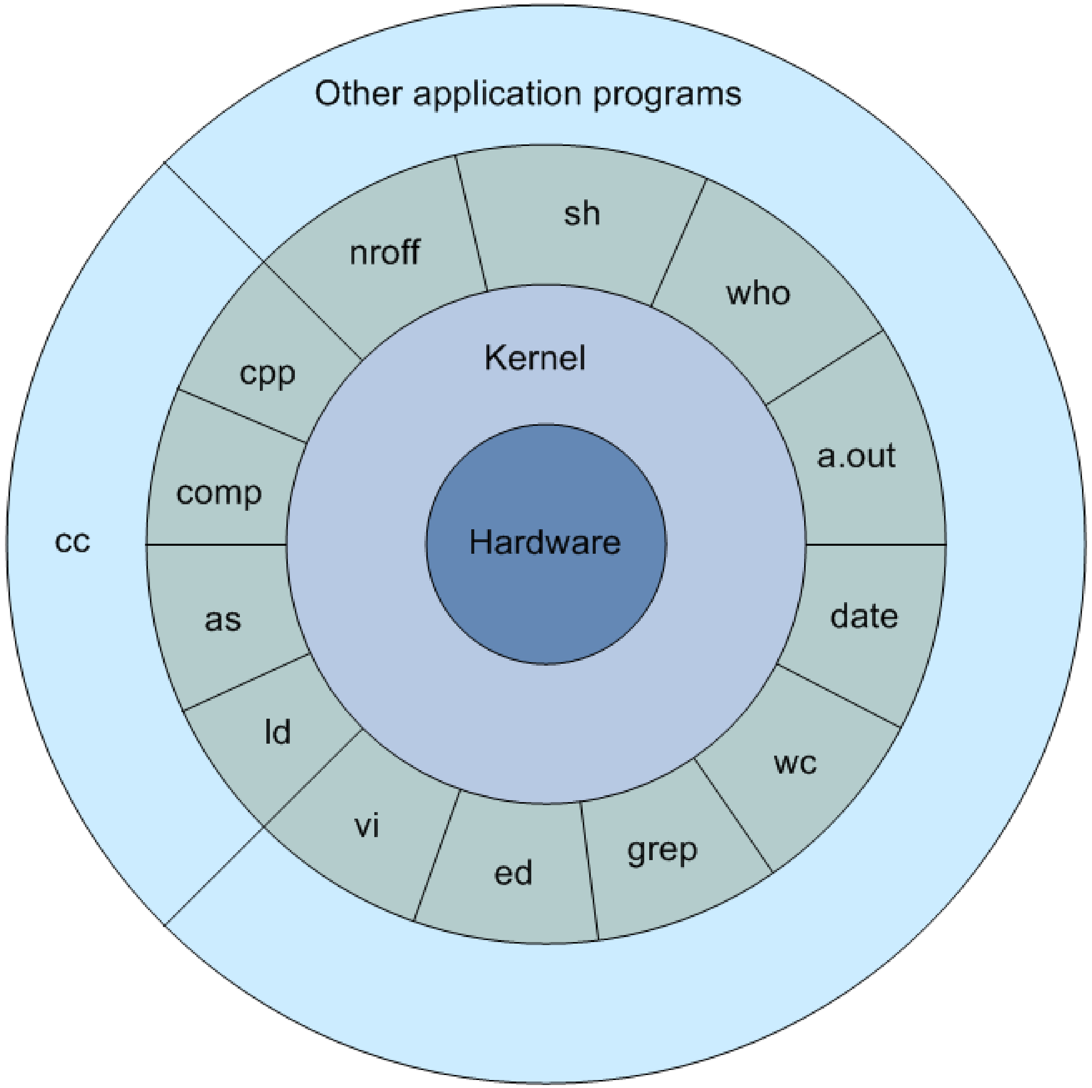


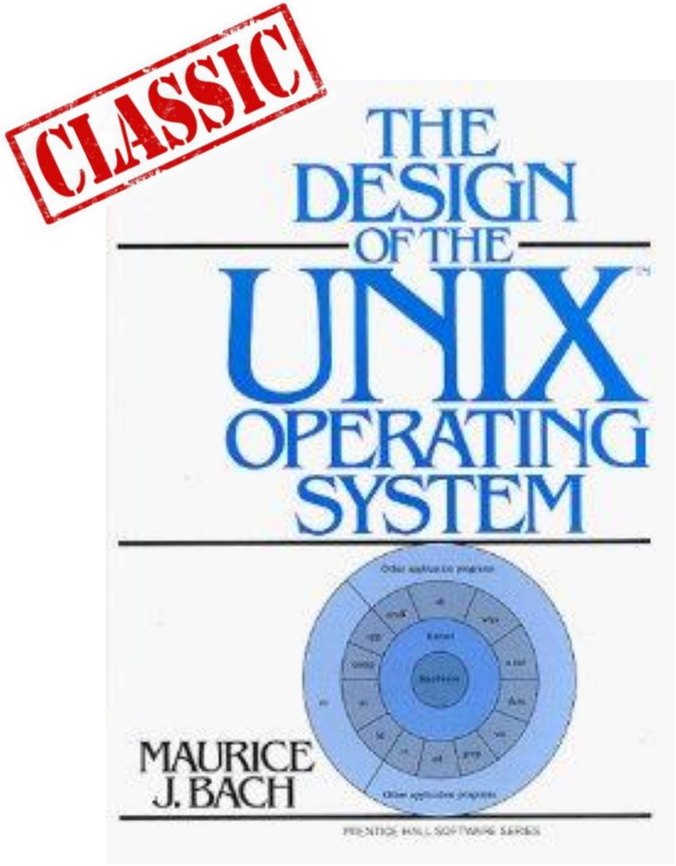
Source: [Clements10]

Layered style – example 2

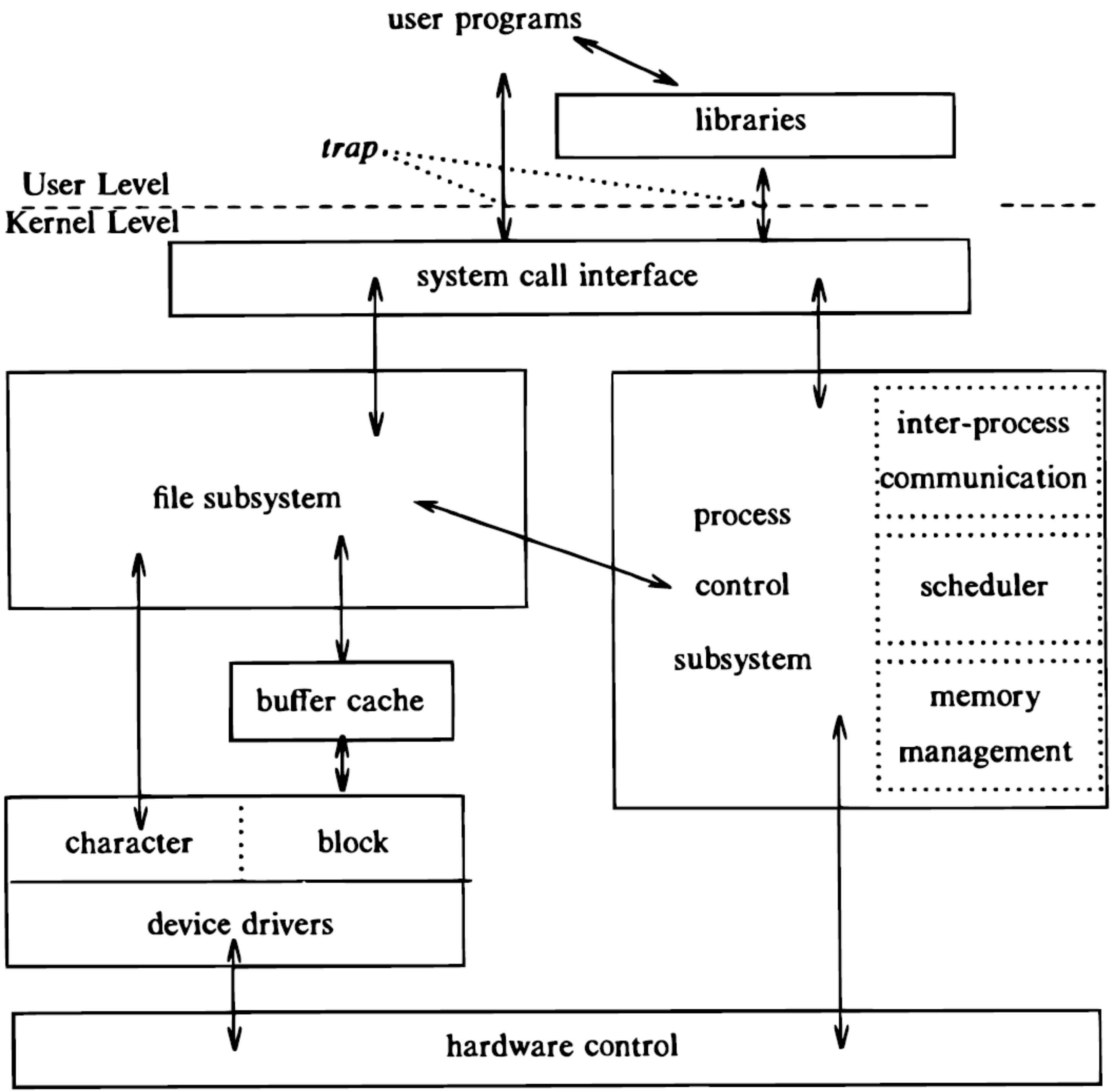


Layered style – example 3

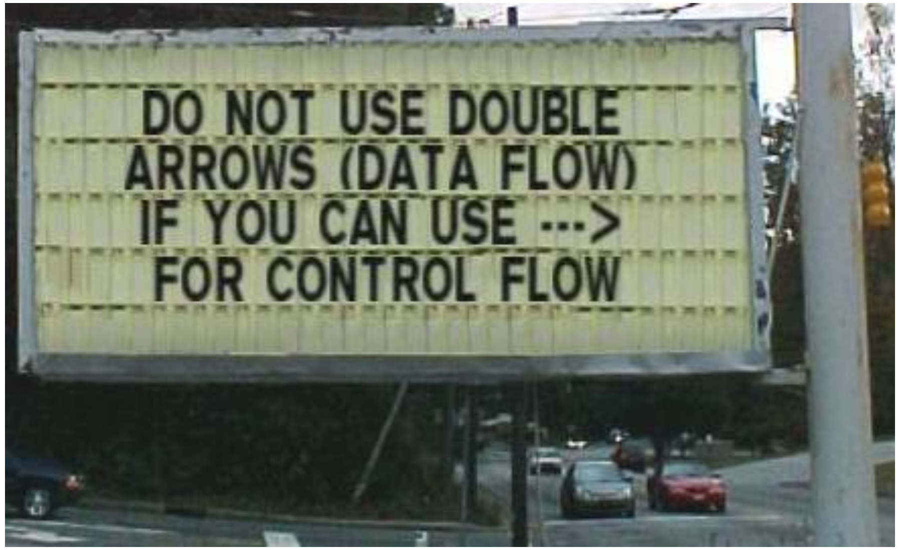




M. Bach. *The Design of the UNIX Operating System.* Prentice Hall, 1986



Layered style – example 4



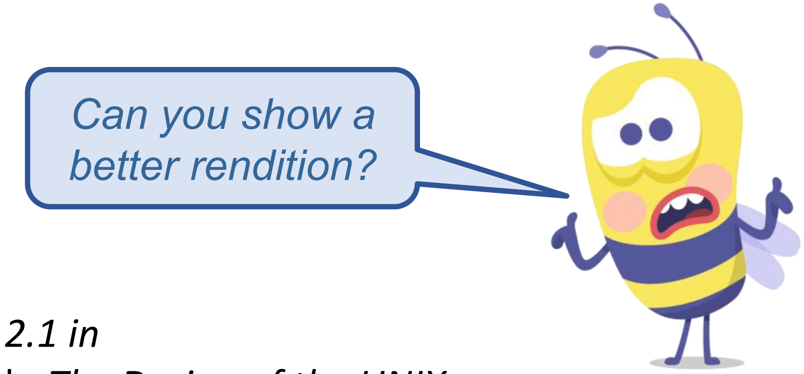


Figure 2.1 in
M. Bach. *The Design of the UNIX Operating System.* Prentice Hall, 1986

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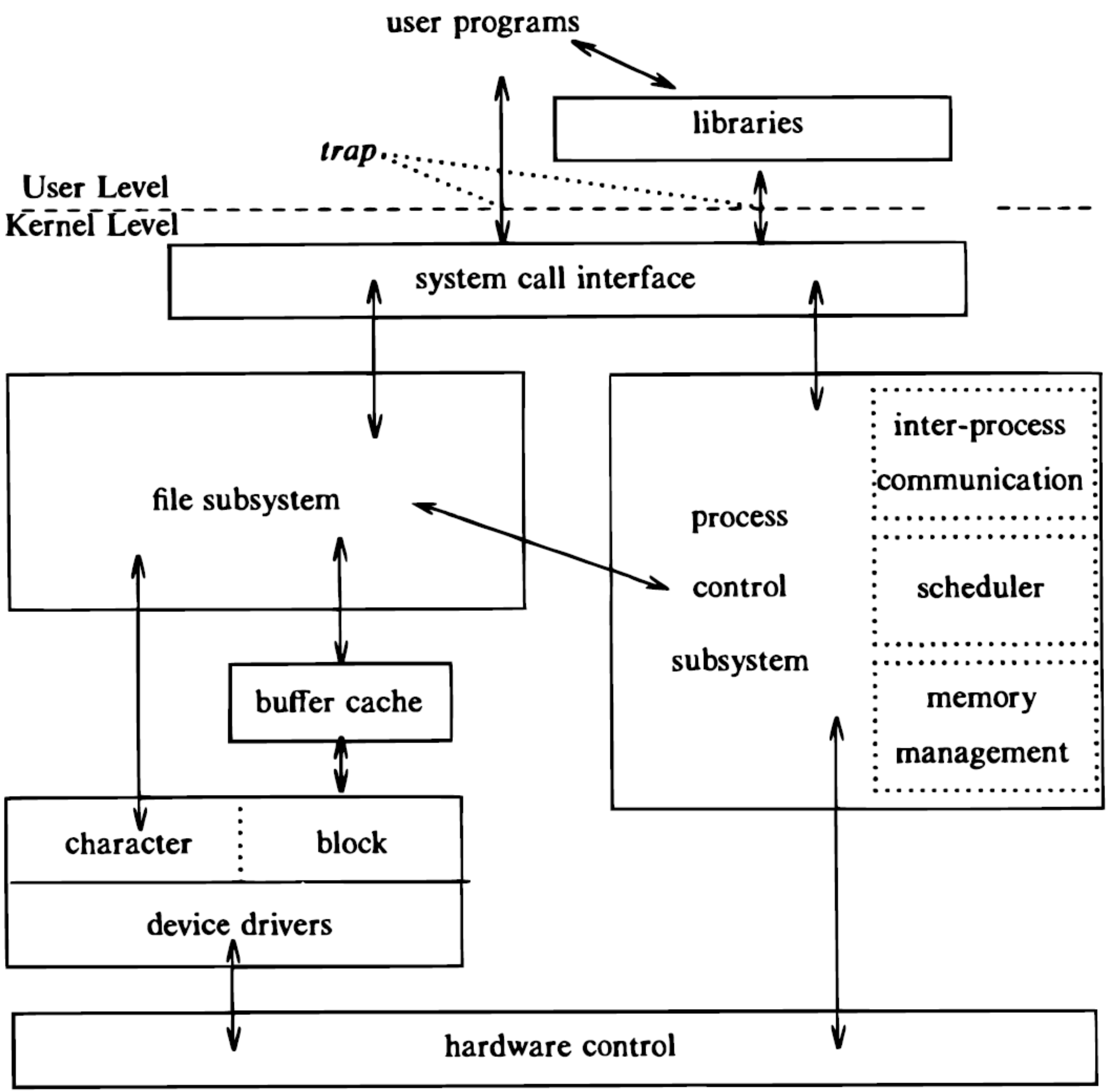
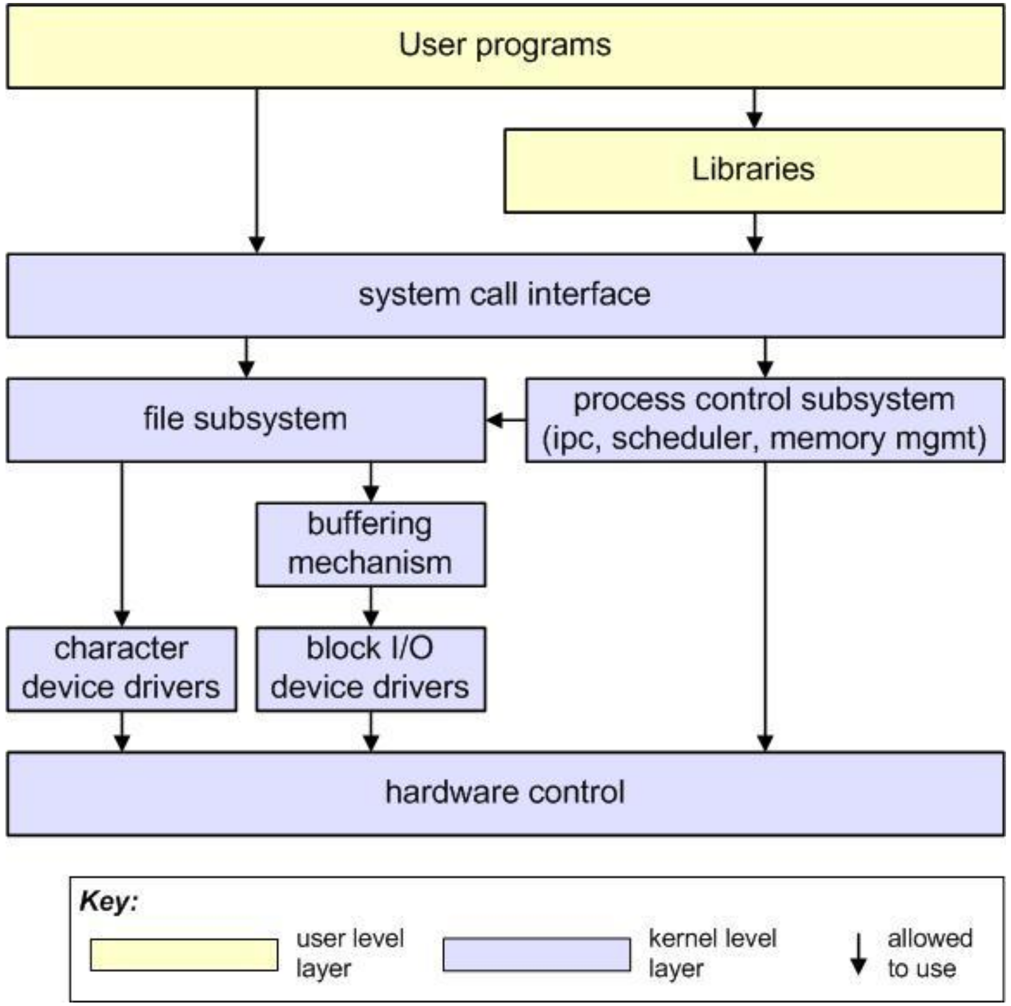

user programs
libraries
trap
User Level
Kernel Level
system call interface
file subsystem
inter-process
communication
process
control
subsystem
scheduler
memory
management
buffer cache
character
block
device drivers
hardware control


User programs
Libraries
system call interface
file subsystem
process control subsystem
(ipc, scheduler, memory mgmt)
buffering
mechanism
character
device drivers
block I/O
device drivers
hardware control
Key:
user level
layer
kernel level
layer
allowed
to use

Questions?

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